Better Forecasts, Higher Trading Profits

Case study: Battery Energy Storage System Trading in Germany

Summary

Battery Energy Storage Systems (“BESS”) are booming around Europe, based on the high expected arbitrage margins achievable by trading BESS assets in different short-term energy and capacity markets.

BESS traders achieve trading margins in several different markets, including the Day-Ahead Auction (DAA), intraday markets, and balancing markets. This paper focuses on the margin achievable in the DAA. The DAA is the largest and most liquid short term power market. Currently in Germany, BESS traders typically count on the DAA for around 20-30% of their annual BESS trading margin. So the DAA is a logical place to start for an analysis of forecast trading value.

As in all markets, in the DAA, accurate forecasting plays an important part in the success of BESS trading. With better forecasts, bids and offers in the DAA markets can be improved in terms of timing, quantity and price.

Yet many BESS optimizers lack a transparent, quantified perspective on *how much* the accuracy of their price forecasts affects their BESS trading results.

To increase transparency, the FlexUp team built a tool to simulate and quantify the incremental trading margin from trading a 1MW / 2MWh BESS in the German DAA, using different forecasts. We compare the trades and resulting margin based on DAA price forecasts from a commercial price forecaster, Electricity Maps with the trades and resulting margin from a “naive” daily Persistence Forecast (24-hour delayed signal: “Today is like yesterday”). We also compare Electricity-Maps -based trading results to the resulting margin from Perfect Foresight (= actual prices). This is done over a 126 study period from 1st October 2025 to 21 Feb 2026.

Better forecasts enable significantly better trading results. Trading based on the forecasts from Electricity Maps improves DAA margin by **€3645 / MW** vs. trading based on the Persistence Forecast, just over these 126 days. The electricity Maps forecast captures 82.7% of the total possible, perfect-foresight gross margin (incl. physical losses).

Methodology

To assess the value of forecasting in BESS trading, our tool runs the same time-arbitrage trading strategy in the DAA with three different forecasts and compares the resulting annual trading margin.

We compare the following three forecasts:

1) **Persistence Forecast**: Vector of prices for the 96 periods in day D = Vector of prices for the 96 periods in day D-1.
2) **Electricity Maps Forecast**: Day ahead forecast published at D-1 11AM, i.e. 1 hour before the DAA gate closure.
3) **Perfect Foresight**: The forecast equals the actual market results, so that the forecasting error equals zero.

Based on this forecast, we simulated a simple time-arbitrage strategy in the Day-Ahead-Auction in the DE-LU zone (basically: buy/charge during low-price periods and sell/discharge during high-price periods, pocket the difference). The strategy is simulated with a single-step linear optimization, which sets a charge/discharge (i.e. buy/sell) schedule, given expected prices, to maximize trading margin.

To account for the effects of forecast error, the optimizer computes the trading decisions using the forecasted prices, while the resulting trading margins are calculated using actual prices.

In addition, the analysis takes into account the effect of out-of-the-money bids and resulting partial/zero fills. This is based on a rational bid and ask price based on the forecast's own price spread:

- Charge bid = average forecast discharge price × RTE. The trader is willing to pay up to 85% of what the stored energy is expected to earn later, reflecting that 15% will be lost to round-trip inefficiency.
- Discharge ask = average forecast charge price / RTE. The trader will not sell for less than the cost of acquiring the energy, adjusted for the fact that more energy was consumed in charging than is delivered in discharging.

The averages are volume-weighted across all MTUs that the optimizer assigns to charge or discharge for that day.

For example, on a day where the optimizer plans to charge at an average forecast price of €63/MWh and discharge at €96/MWh:

- Charge bid = €96 × 0.85 = €82 (willing to pay up to €82, vs. forecast of €63)
- Discharge ask = €63 / 0.85 = €74 (willing to sell down to €74, vs. forecast of €96)

Based on these bids and asks, for each MTU, the model verifies whether the trader's bid or ask price would have been accepted given the actual clearing price. Out-of-the-money orders are rejected. In the perfect foresight scenario, all bids / asks are in the money.

Subsequently, any rejected bid/ask volume is paired with the corresponding bid/aks, and both buy and sell order are cancelled, and the charge/discharge is not done, thereby missing the

arbitrage opportunity for that cycle. In this way, out-of-the-money bids, due to forecast error, cause missed revenue.

To compare the impact of forecasting errors, we run the daily trading algorithm three times—once with Persistence Forecast, once with Electricity Maps, once with Perfect Foresight. Then we sum up the days and compare the annual trading margin delta per MW.

Trading simulation inputs

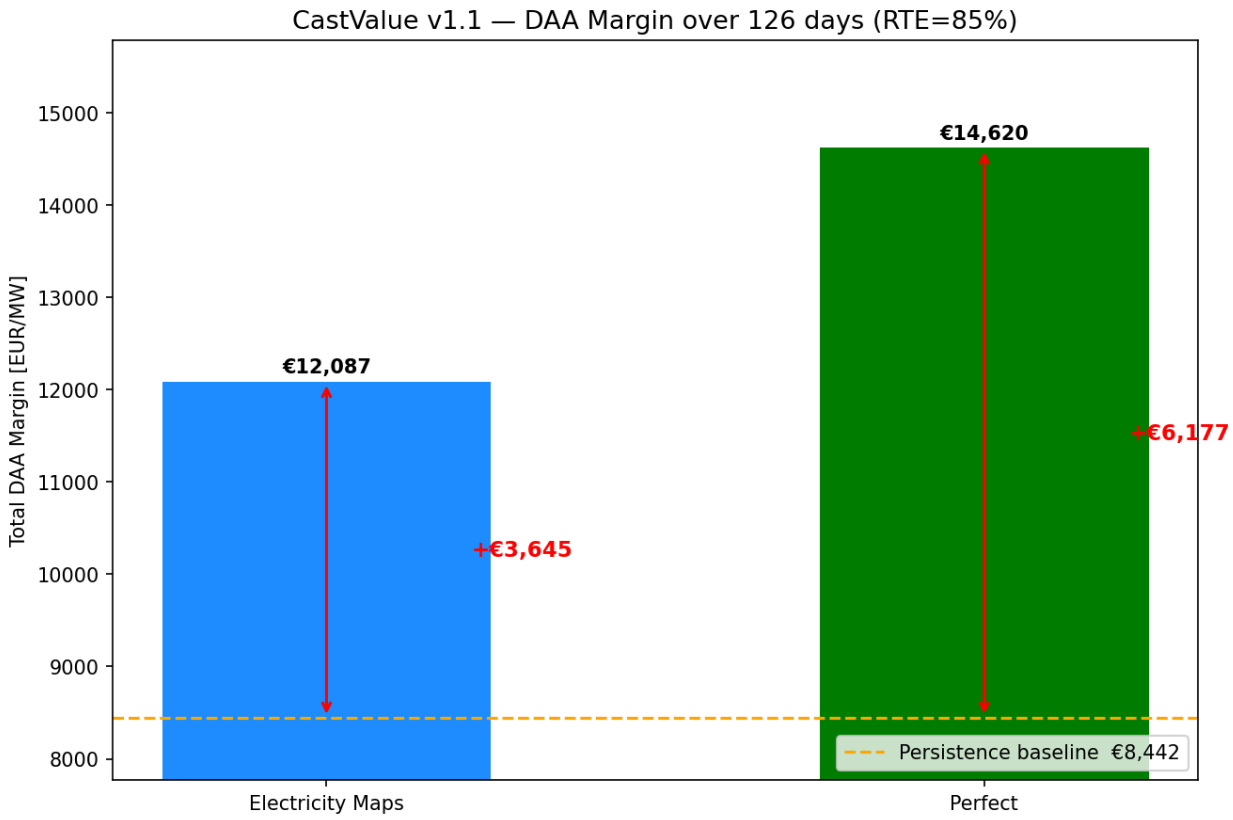
The table below lists the assumptions used for each of the trading scenarios. In summary, each scenario uses the same technical constraints, while varying the DAA price expectations.

		Persistence Forecast	**Electricity Maps Forecast**	**Perfect Foresight**
1	Power capacity	1.0 MW		
2	Energy capacity	2.0 MWh		
3	SOC range	0%-100%		
4	Max. cycles per day	2.0		
5	Round Trip Efficiency	85%		
6	Starting / Ending SOC	Starts at zero on day 1, then end-of-day SOC carries into the next day as start-of-day SOC.		
7	DAA price forecasts used for charge/discharge decisions	Persistence Forecast	Electricity Maps Forecast D-1, 11am	Actual prices

Physical effects such as SOC-measurement errors, line losses are not modelled.

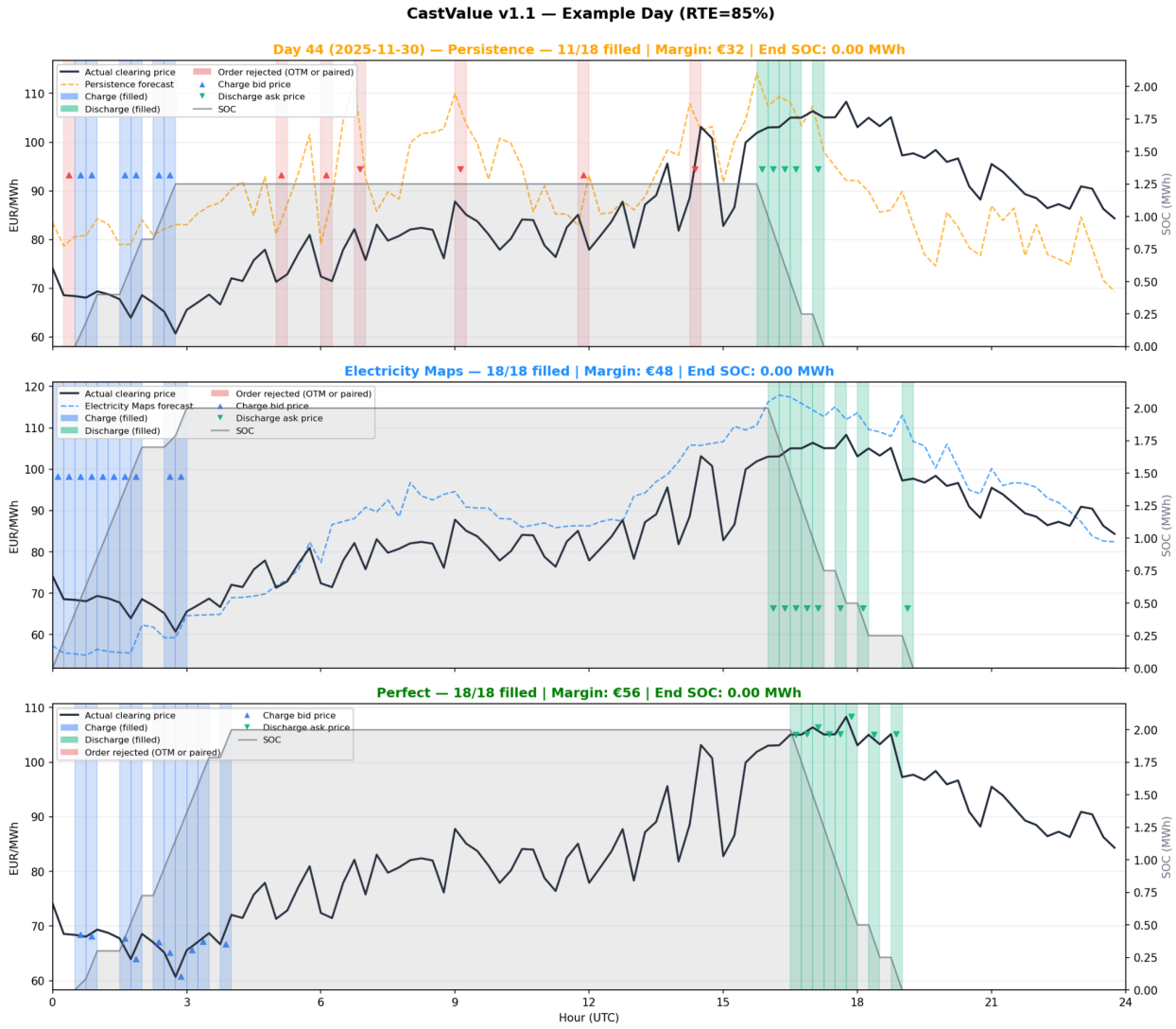
Results

The graph below shows the resulting difference to the Persistence Baseline in trading margin from the DAA market only, depending on the price forecasts used, during the evaluation period.


CastValue v1.1 — DAA Margin over 126 days (RTE=85%)
Total DAA Margin [EUR/MW]
15000
14000
13000
12000
11000
10000
9000
8000
€12,087
+€3,645
€14,620
+€6,177
Persistence baseline €8,442
Electricity Maps
Perfect

Example Day Evaluation

A deep dive in an example day shows clearly how inaccurate price forecasts lead to suboptimal charge/discharge decisions and lower daily arbitrage margins for the BESS trader.



A detailed evaluation of the trading decisions of each of the three “traders” shows how imperfect forecasts lead to suboptimal decisions and lost trading margin:

1. **"Persistence" Trader:** Using yesterday's prices as today's forecast, this trader expects prices about €5/MWh higher than they actually are (forecast avg €90 vs. actual €85, MAE €12.3). The forecast shape is also off — instead of one clean evening peak, persistence spreads its discharge hours across midday (9–17), misreading when the day's premium actually sits. Charge bids sit at €93 and discharge asks at €94, a

razor-thin spread that reflects the flat forecast profile and leaves little protection against error. Seven of the eighteen planned MTUs are rejected (out-of-the-money or paired), so only the luckiest-timed trades survive. Daily profit: €31.79, 61% fill rate, 57% of optimal.

2. **"Provider Forecast" Trader:** The Electricity Maps forecast tracks actual prices closely (MAE €7.70, much better than the Persistence forecast). The trader correctly identifies the cheapest overnight window for charging (hours 0–2) and the evening peak for discharging (hours 16–19). The charge bid of €98 is safely above clearing prices in those hours, and the discharge ask of €66 is comfortably below the evening clearing prices — so all 18 planned MTUs fill. The trader ends the day fully discharged with a daily profit of €47.90, a respectable 85% of optimal, with a 100% fill rate.
3. **"Perfect Foresight" Trader:** With perfect knowledge, the trader charges in the four cheapest slots (hours 0–3) and discharges in the three highest-priced slots (hours 16–18), bidding exactly at the clearing price to guarantee every fill. Crucially, the RTE physical loss is still paid — 2.35 MWh bought to deliver 2 MWh sold. Despite this physical friction, with perfect scheduling and bidding, the trader achieves daily profit of €56.26 (100%), with a 100% fill rate.

In short, this example shows that differences in forecast accuracy can have a significant impact on trading margins and - given this bidding strategy - on fill rates. Better forecasts enable traders/algos to outperform their peers and approach optimal results for themselves and their clients.

Limitations and extensions

To show the basic dynamics of how better forecasting drives BESS trading results, this analysis is purposely quite simple. As such, it is important to point out some of the main limitations of this "v1.0" analysis, mainly that this analysis only covers physical time-arbitrage within the DAA, and not several other markets (such as intraday-auction, intraday continuous or ancillary services), nor other trading opportunities (such as cross-market arbitrage (e.g. DAA vs. IDC), real-option-based strategies ("virtual cycling"), or any other strategies. Many more arbitrage opportunities exist for a BESS that is traded cross-market in Germany. Improved forecasting is key to outperforming peers in all of these markets.

Given these limitations, FlexUp will continue to work on the forecast value tool, specifically focusing on the following extensions / improvements:

- Covering additional markets, such as intraday and reserve markets
- Replicating the analysis for other geographies, with different general forecasting accuracy levels. Germany tends to have high forecast errors given the renewable share)
- Comparing additional forecasters beyond Electricity Maps to benchmark forecasters against their peers, not just a naive Persistence Forecast
- Replicating the analysis for different trading contexts, including cross-border / connector arbitrage.

These interesting extensions will further generate transparency in price forecasting and power markets.

Further information

This whitepaper is a collaboration between FlexUp GmbH and Electricity Maps ApS.

FlexUp GmbH operates the Teamcast collaborative forecasting platform. This platform collects, compares and ensembles day-ahead price forecasts from many different providers in order to create transparency in forecasting performance and use the “wisdom of the crowd” to build better day ahead price forecasts. Forecast accuracy is a key factor in keeping electricity markets functioning smoothly, reducing price volatility and making the energy transition affordable in Europe. For further information contact: hello@flexup.pro.

Electricity Maps has built **t**he world’s most comprehensive electricity data platform. Electricity Maps provides global access to electricity mix, prices, load, and carbon intensity. Data is available in real-time, historically and forecasted. Electricity Maps gives you unprecedented insights into the grid, with seamless access to global data at the highest granularity. For further information visit https://www.electricitymaps.com/ or get in touch here: https://www.electricitymaps.com/contact